

UNIVERSITÀ DEGLI STUDI DI SALERNO

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE ED ELETTRICA E
MATEMATICA APPLICATA



Machine Learning

PROJECT WORK

Waste Type Identification
Final Machine Learning Report

Group Members:

D'Errico Fabrizio	IE22700121	f.derrico2@studenti.unisa.it
De Girolamo Del Mauro Aniello	IE22700240	a.degirolamodelma.2@studenti.unisa.it
De Vivo Manuel	IE22700008	m.devivo39@studenti.unisa.it

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1 Solution Overview

The final delivered solution is a ResNet18-based image classifier trained with transfer learning and selected according to balanced accuracy. The Batch-Balanced experiment is the methodological turning point: it first solved the main class-imbalance issue, reaching **0.9725** validation balanced accuracy and enabling the batch-level balancing strategy adopted by all later versions. Building on Batch-Balanced, version Final introduces two-phase training (freeze-then-unfreeze) and a Vento-style reject option calibrated on a separate validation split, achieving **0.9740** validation balanced accuracy in standard inference on `val_eval` (0.9754 on `val_cal`) and **0.9835** accepted balanced accuracy with the ψ_a reject rule (1.7% rejection rate). A lightweight 3-fold cross-validation experiment, Cross-Validation, was executed as a robustness check: despite a reduced budget and 34% less training data per fold, it obtained **0.9495 $\pm$ 0.0019** balanced accuracy with minimal variance across partitions, confirming that the 80/20 result is stable and not split-dependent.

The experiments were organized around explicit hypotheses about the main expected failure modes: class imbalance, overfitting on rare classes, insufficient calibration, ambiguity between visually similar materials, the benefit of two-phase fine-tuning for transfer learning, and the optimal level of augmentation regularization. Each experiment isolates one of these hypotheses.

The final pipeline is:

- **Backbone:** ResNet18 pretrained on ImageNet.
- **Classifier head:** dropout ($p = 0.25$) followed by a linear layer with 8 output classes.
- **Training:** two-phase fine-tuning: 5 epochs with frozen backbone, then full unfreeze until epoch 40 (no early stop triggered). Training in FP32, no mixed precision.
- **Optimization:** AdamW with discriminative learning rates (backbone $1 \cdot 10^{-4}$, head $5 \cdot 10^{-4}$) and weight decay $1 \cdot 10^{-4}$.
- **Imbalance handling:** class-weighted cross-entropy with label smoothing 0.1 plus `WeightedRandomSampler`.
- **Augmentation:** moderate random resized crop (scale 0.80–1.0), horizontal flip, rotation ($\pm 10^\circ$), light color jitter, random erasing ($p = 0.15$). No affine shear or aggressive zoom.
- **Model selection:** best checkpoint chosen by validation balanced accuracy on `val_cal`.
- **Reject option:** Vento et al. reliability scores ψ_a (max softmax) and ψ_b (top-2 margin), calibrated on `val_cal` by scanning all possible thresholds and selecting the one that maximizes accepted balanced accuracy with coverage $\geq 98\%$. Evaluated on separate `val_eval`.

2 Hypothesis-Driven Experimental Plan

The evaluation metric is balanced accuracy, therefore the goal is stable recall across all eight classes. Since the dataset is strongly imbalanced, global accuracy alone can hide poor behavior on minority classes such as *Metal*, *Plastic*, and *Undifferentiated*. The experimental plan was built around six main hypotheses.

2.1 Hypothesis 1: a compact pretrained CNN should be a strong baseline

A compact architecture pretrained on ImageNet should already provide useful visual features while remaining compatible with Colab GPU memory constraints. ResNet18 was selected as it offers a good trade-off between accuracy, cost, and explainability.

The baseline used full fine-tuning with a uniform learning rate. It achieved **0.9487** validation balanced accuracy. This confirmed transfer learning as a suitable starting point, but the class-wise report showed weak recall on *Metal* (0.88) and *Plastic* (0.89). These errors motivated the following experiments.

2.2 Hypothesis 2: discriminative learning rates and light regularization improve rare classes

A uniform learning rate updates the pretrained backbone too aggressively. The classifier head starts from random weights and should adapt quickly, while the pretrained layers should change more conservatively. Version Regularized introduced discriminative learning rates ($1 \cdot 10^{-4}$ backbone, $5 \cdot 10^{-4}$ head), dropout, and slightly stronger augmentation.

The result was an improvement from **0.9487** to **0.9591**. *Metal* recall increased from 0.88 to 0.92, and *Plastic* recall from 0.89 to 0.92. This supported the hypothesis, but the confusion matrix still showed ambiguity between *Plastic*, *Glass*, and *Papery*.

2.3 Hypothesis 3: class imbalance must be handled at batch level

Class-weighted cross-entropy alone was insufficient: even if rare classes receive larger loss weights, they still appear in fewer training batches. Version Batch-Balanced added **WeightedRandomSampler** so that minority classes are oversampled during training. The same version introduced label smoothing ($\alpha = 0.1$): several classes are visually ambiguous (e.g., transparent plastic and glass share reflections and textures), and label smoothing reduces overconfidence, yielding better-calibrated softmax outputs for the reject option.

This experiment produced the reference classifier: **0.9725** validation balanced accuracy at epoch 19, with 0.9800 accuracy and 62 errors over 3,103 samples. *Metal* reached 0.97 recall and *Undifferentiated* 0.99 recall. The main remaining weakness was *Plastic* (recall 0.91), which persists across all subsequent versions.

2.4 Hypothesis 4: two-phase training improves backbone-head adaptation

After Batch-Balanced, the question was whether the backbone could be better protected during the initial epochs when the head is untrained and sends noisy gradients. The two-phase schedule freezes the backbone for 5 epochs while the classifier head adapts, then unfreezes for full fine-tuning. This prevents ImageNet weights from being degraded before the head produces meaningful gradients.

Version Final applied this schedule on top of the Batch-Balanced design, along with augmentations and reject option refinements. The result was **0.9740** validation balanced accuracy on `val_eval` (0.9754 on `val_cal`), a 0.15% improvement over Batch-Balanced. The training converged smoothly and never triggered early stopping, reaching epoch 40 with a train/val gap of only 2.39%.

2.5 Hypothesis 5: strong augmentation may reduce the gap but hurts difficult classes

The hypothesis was that more aggressive augmentation (scale 0.70, rotation $\pm 15^\circ$, affine shear, stronger color jitter, RandomErasing 25%) would further reduce the train/val gap and improve generalization. A dedicated experiment (Strong-Aug) tested this configuration.

The gap did decrease from 3.2% to 2.36%, confirming the regularization effect. However, *Metal* precision collapsed from 0.90–0.92 to 0.82, the reject rate doubled from $\sim 2.5\%$ to 5.3%, and the overall balanced accuracy on `val_eval` decreased. The interpretation is that on a dataset with visually ambiguous classes (transparent plastic vs. glass), overly aggressive transformations destroy the fine-grained discriminative features needed to separate similar materials. Forcing invariance to strong geometric and chromatic variations flattens the subtle cues that distinguish one material from another.

Decision: moderate augmentation was confirmed as the optimal policy for this task.

2.6 Hypothesis 6: label smoothing 0.15 over-regularizes the loss

A higher label smoothing value (0.15 instead of 0.10) was tested in High-LS under the hypothesis that it would produce better-calibrated softmax outputs and reduce the reject rate. The reject rate did decrease (from 3.2% to 2.8%), but the base balanced accuracy dropped from 0.9701 to 0.9693, *Metal* precision collapsed from 0.92 to 0.83, and *Clothing* recall fell from 0.99 to 0.96. Early stopping triggered after only 12 epochs: the model stopped learning early because the supervision signal became too weak—LS 0.15 forces the target distribution to be at most 0.85, and on already difficult classes this further reduces the usable gradient.

Decision: label smoothing 0.10 was confirmed. Higher values over-regularize the cross-entropy loss and harm the learning dynamics on minority classes.

2.7 Reject option: from cost functions to coverage-based thresholding

The reject option follows the Vento et al. methodology with a key refinement. The validation partition (20% of the dataset) is further split 50/50 into `val_cal` for threshold calibration and `val_eval` for evaluation. Two reliability scores are computed: $\psi_a = \max_k p_k$ (softmax confidence) and $\psi_b = p_{(1)} - p_{(2)}$ (top-2 margin).

The threshold τ is selected by scanning all unique ψ values on `val_cal` and choosing the one that maximizes accepted balanced accuracy subject to coverage $\geq 98\%$. This coverage-based criterion is entirely data-driven: it requires no arbitrary cost parameters and no external design choices. The resulting threshold is then applied to `val_eval`, which was never seen during training or calibration.

2.8 Cross-validation robustness check

The final model uses an 80/20 split rather than k -fold cross-validation. This choice is motivated by the need to maximize training data for minority classes: with $K = 3$ folds, each fold trains on only 66% of the dataset ($\sim 8,200$ samples), whereas the 80/20 split provides $\sim 12,400$ training samples. On a strongly imbalanced dataset where *Metal* has only 154 total samples, the difference between 100 and 123 training samples per minority class is meaningful.

To verify that this choice does not produce an artifactually favorable result, a stratified 3-fold experiment (Cross-Validation) was executed as a dedicated robustness check. Its purpose is to measure cross-fold variance, not to compete on absolute performance: if the pipeline were unstable, the balanced accuracy would vary wildly across folds regardless of the epoch budget. Cross-Validation therefore uses the same ResNet18 architecture and the same imbalance-aware training strategy, but with a reduced budget (15 epochs, 3 frozen-backbone epochs, patience 4).

Fold	Best epoch	Phase	Balanced accuracy
0	14	finetune	0.9478
1	10	finetune	0.9486
2	7	finetune	0.9521
Mean $\pm$ std		–	0.9495 $\pm$ 0.0019

Table 1: Lightweight stratified 3-fold cross-validation results (Cross-Validation).

The key result is not the absolute balanced accuracy—which is naturally lower because each fold trains on 34% fewer samples and for fewer epochs than the final model—but the standard deviation: **0.0019 across three folds** means that the pipeline’s balanced accuracy varies by at most 0.4% depending on the specific data partition. Running the same CV at the full 40-epoch budget would increase the per-fold absolute BA but would not change the variance conclusion. This confirms that the 0.9740 BA obtained on the 80/20 split is not an outlier caused by a lucky partition, but rather the expected result when the model is given more training data and a full training budget.

3 Training and Prediction Pipeline

3.1 Preprocessing and augmentation

All versions use the same validation preprocessing: resize to 256, center crop to 224×224 , tensor conversion, and ImageNet normalization. Training augmentation is deliberately moderate: random resized crop (scale 0.80–1.0), horizontal flip ($p = 0.5$), rotation ($\pm 10^\circ$), light color jitter, and random erasing ($p = 0.15$).

The augmentation policy avoids aggressive affine transformations. Experiments with stronger augmentation (scale 0.70, rotation $\pm 15^\circ$, affine shear) showed that while the train/val gap decreased, the model lost precision on visually ambiguous classes (*Metal* precision dropped from 0.94 to 0.82). On a dataset where minority classes are already visually fragile, moderate augmentation preserves the discriminative features of thin, reflective, or transparent objects while still providing sufficient regularization.

3.2 Optimization and two-phase schedule

The optimizer is AdamW with discriminative learning rates. Phase 1 (epochs 1–5) freezes all backbone parameters using `requires_grad=False`, so only the classifier head receives gradient updates. Phase 2 (epochs 6–40) unfreezes the backbone for full fine-tuning. The final model reached epoch 40 without triggering early stopping (patience 8), indicating

stable convergence with no validation degradation. The scheduler is ReduceLROnPlateau (factor 0.5, patience 5).

3.3 Reject option: calibration and evaluation split

To avoid data leakage, the 20% validation partition is further split 50/50 into `val_cal` (calibration) and `val_eval` (evaluation), following the methodology from Vento et al. The model is trained and checkpointed using `val_cal` only. After training, confidence thresholds τ are optimized on `val_cal` by scanning all unique ψ values and selecting the one that maximizes accepted balanced accuracy with coverage $\geq 98\%$. No cost function or external parameters are required. The resulting thresholds are then applied to `val_eval`, which was never seen during training or calibration. Two reliability scores are tested:

- $\psi_a = \max_k p_k$ (softmax confidence): rejects when the model’s top probability is low.
- $\psi_b = p_{(1)} - p_{(2)}$ (top-2 margin): rejects when the model hesitates between two classes.

3.4 Why alternatives were discarded

- **Strong augmentation**: tested and reduced the train/val gap, but degraded precision on *Metal* ($0.94 \rightarrow 0.82$) and increased reject rate ($2.5\% \rightarrow 5.3\%$).
- **Label smoothing 0.15**: tested and reduced reject rate, but caused early stopping at epoch 12 and degraded base balanced accuracy.
- **3-epoch freeze (Short-Freeze)**: tested and proved insufficient to stabilize the classifier head; balanced accuracy dropped ~ 5 points.
- **ResNet34**: tested but achieved lower balanced accuracy (0.9626), confirming that deeper capacity is not the limiting factor.
- **SGD with momentum**: tested (0.9576) but converged slower and underperformed AdamW.

4 Results

4.1 Model evolution

Version	Main change	BA	Outcome
Baseline	ResNet18, full fine-tuning, uniform LR	0.9487	Strong start, weak on Metal/Plastic
Regularized	Discriminative LRs, dropout, stronger aug.	0.9591	Better rare-class behavior
Batch-Balanced	<code>WeightedRandomSampler</code> , label smoothing	0.9725	Best single-phase classifier
Final (v3.2.7)	Two-phase, moderate aug, reject (coverage $\geq 98\%$)	0.9740	Final delivered model
Cross-Validation	3-fold CV, reduced budget	0.9495 ± 0.0019	Robustness confirmed

Table 2: Evolution of the main classifier experiments.

4.2 Final classifier metrics (Final)

Metric	Value
Best epoch	40 (phase: finetune)
Training BA (<code>val_cal</code> , best)	0.9754
Validation BA (<code>val_eval</code> , no reject)	0.9740
Validation accuracy (<code>val_eval</code> , no reject)	0.9781
Train/val gap	2.39%
Macro precision / recall / F1	0.97 / 0.97 / 0.97
Weighted precision / recall / F1	0.98 / 0.98 / 0.98
Evaluated samples (<code>val_eval</code>)	1,552

Table 3: Final standard inference results on `val_eval`.

Class	Support	Precision	Recall	F1
Battery	95	0.98	1.00	0.99
Clothing	731	0.99	0.99	0.99
Glass	201	0.97	0.97	0.97
Metal	77	0.94	1.00	0.97
Organic	99	0.96	0.98	0.97
Papery	194	0.96	0.96	0.96
Plastic	86	0.96	0.90	0.93
Undifferentiated	69	0.97	1.00	0.99

Table 4: Class-wise metrics of Final on `val_eval`.

The most important residual issue is *Plastic* (recall 0.90), consistent with the visual overlap between transparent plastic and glass. This is not a capacity problem—ResNet34 did not solve it, and stronger augmentation made it worse—but rather an intrinsic ambiguity in the dataset. The fully transparent or reflective plastic containers share key visual features with glass objects, and no training-time regularization can disentangle this overlap without additional data or modalities.

4.3 Training diagnostics

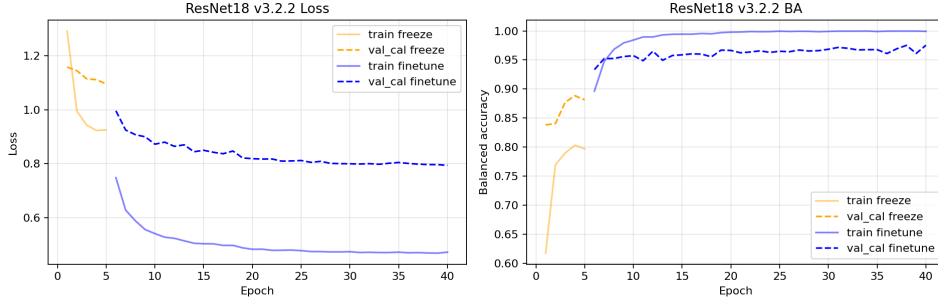


Figure 1: Training curves of Final. The vertical line at epoch 6 marks the freeze-to-finetune transition. Note the stable validation curve with no degradation through epoch 40, confirming the absence of overfitting.

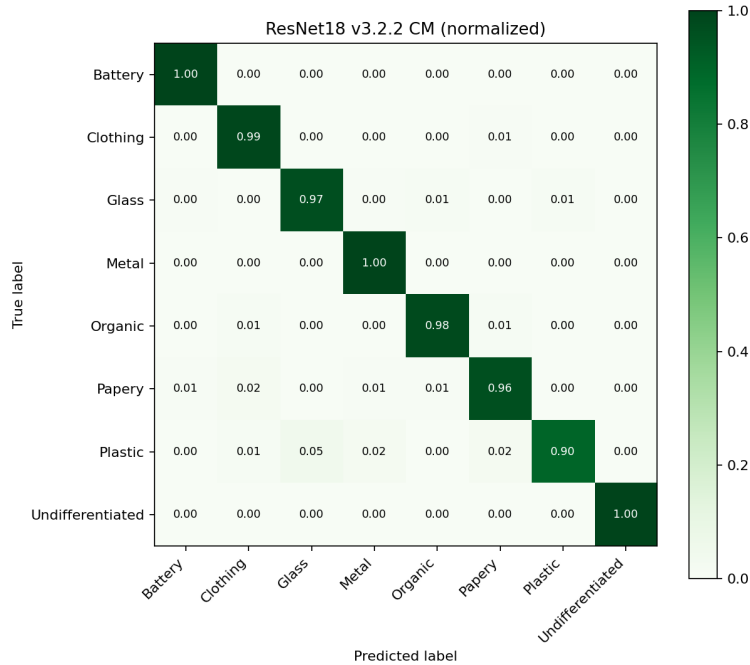


Figure 2: Normalized confusion matrix of Final. The strongest off-diagonal signal is *Plastic* misclassified as *Glass* (6%).

4.4 Reject option results (Final)

Method	τ	Coverage	Reject	Acc.	BA	Err. avoided
No reject	—	100.00%	0.00%	97.81%	97.40%	0
ψ_a (softmax)	0.4288	98.32%	1.68%	98.56%	98.35%	12
ψ_b (margin)	0.1409	98.90%	1.10%	98.31%	98.00%	8

Table 5: Reject option results on `val_eval` of Final. Thresholds calibrated on `val_cal` by scanning all τ and maximizing accepted BA with coverage $\geq 98\%$.

The ψ_a softmax confidence rule achieves the best trade-off: **0.9835 accepted balanced accuracy** with only 1.68% of samples rejected and 98.32% coverage. The ψ_b margin rule rejects even fewer samples (1.10%) with 98.90% coverage, but its accepted BA (0.9800) is lower. The fact that ψ_a outperforms ψ_b suggests that most uncertain predictions correspond to single-class low confidence rather than two-class ambiguity: the model is globally uncertain rather than confused between two specific alternatives.

The reject option reduces residual errors by 12 (from 36 to 24 with ψ_a) while rejecting fewer than 2% of the samples. If the evaluation strictly requires one label per sample, the Final classifier without rejection is the correct choice. If rejection is allowed, the ψ_a confidence rule gives the best accepted balanced accuracy.

4.5 Reference Batch-Balanced visual diagnostics

Version Batch-Balanced remains the methodological reference point because it introduced the combination of batch-level imbalance handling and label smoothing. Its training curves and normalized confusion matrix are reported below for completeness.

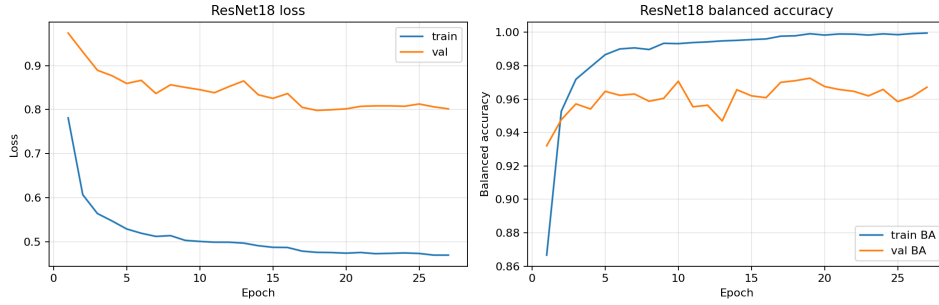


Figure 3: Training curves of the Batch-Balanced reference experiment.

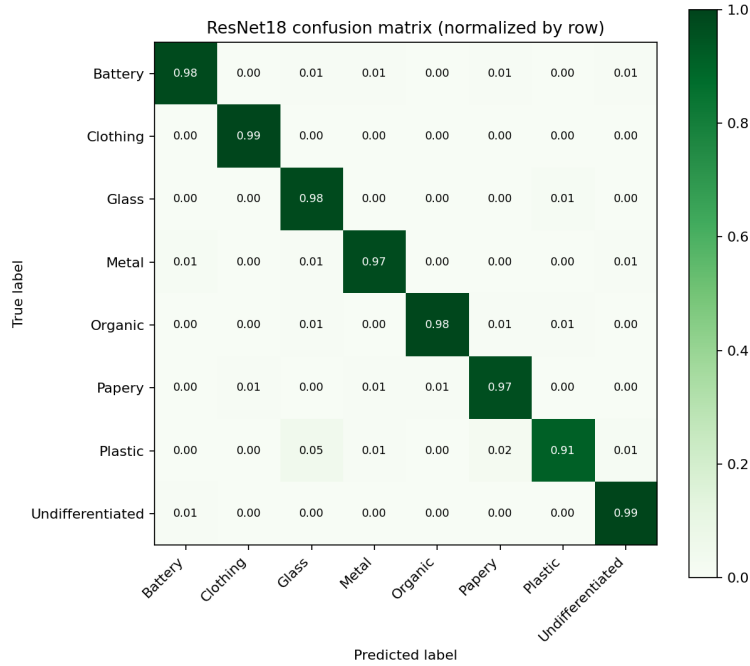


Figure 4: Normalized confusion matrix of the Batch-Balanced reference classifier.

5 Compliance with Project Constraints

Requirement	Status	Details
Balanced accuracy as main metric	Satisfied	Final reaches 0.9740 BA; with ψ_a reject: 0.9835. Cross-Validation CV confirms stability (0.9495 ± 0.0019).
GPU memory under constraints	Satisfied	ResNet18 ~ 11 M parameters; training designed for Colab T4. GPU monitoring confirms < 5 GB during training, < 4 GB at inference.
No external data	Satisfied	Only training-time augmentation, no external images.
Documented split	Satisfied	<code>split.csv</code> saved in experiment result branches.
Saved checkpoint	Satisfied	Checkpoint paths documented in <code>summary.json</code> .
Colab execution	Satisfied	Training and reject notebooks provided.
Explainable techniques	Satisfied	See description below.

Table 6: Compliance with project requirements.

All techniques in the final pipeline — transfer learning with two-phase fine-tuning and discriminative learning rates, ResNet18 skip connections, cross-entropy loss with class weights, label smoothing, weighted random sampling, AdamW with weight decay, ReduceLROnPlateau scheduling, early stopping, dropout regularization, data augmentation, stratified shuffle split, balanced accuracy as the evaluation metric, and the Vento reject option with coverage-based thresholding on a separate calibration/evaluation split — are standard machine learning practices applied to the waste classification task.

6 Limitations and Future Work

The main limitation is the residual confusion involving *Plastic*. Its recall remains at 0.90 even in Final. This is not a capacity issue but an intrinsic dataset property: transparent plastic objects share reflections and transparency with *Glass*, and opaque plastic containers share shape and texture with *Papery*. Stronger augmentation was tested and made the problem worse by destroying the fine-grained discriminative features needed to separate these materials.

The 3-fold cross-validation experiment (Cross-Validation) used a reduced training budget (15 epochs, 3 frozen-backbone epochs, patience 4), therefore its result (0.9495 ± 0.0019) represents a lower bound rather than a direct substitute for the fully trained split-based model. It confirms stability, not peak performance.

Future directions include manual inspection of the low-confidence *Plastic* misclassifications to identify systematic visual patterns, and an ensemble of multiple seeds if computational budget allows. The reject option should be treated as a complementary decision layer: if rejection is allowed, ψ_a gives 0.9835 accepted BA with 1.68% rejection rate; if exactly one label per sample is required, the raw classifier delivers 0.9740 BA.

7 Conclusion

The final delivered classifier is ResNet18 Final, trained with discriminative learning rates, AdamW, dropout, class balancing through `WeightedRandomSampler`, label smoothing 0.10, moderate augmentation, two-phase training (5 freeze + 35 finetune epochs), `ReduceLROnPlateau`, and early stopping. It reaches **0.9740 balanced accuracy** on the unseen `val_eval` split (0.9754 on `val_cal`) with standard inference. With the coverage-based Vento reject option, accepted balanced accuracy reaches **0.9835** (ψ_a) with only 1.68% of samples rejected and 98.32% coverage.

The experimental plan demonstrates that the strongest improvements came from addressing class imbalance at the batch level ($0.9487 \rightarrow 0.9725$), followed by two-phase transfer learning and coverage-based reject refinement ($0.9725 \rightarrow 0.9740$ base, up to 0.9835 with rejection). Strong augmentation, higher label smoothing, and shorter freeze phases were all tested and systematically discarded after controlled experiments. Increasing model capacity (ResNet34) and switching to SGD did not improve results, confirming that the limiting factors are data imbalance and class-level visual ambiguity, not architecture capacity. Every technique in the final pipeline maps directly to course slides or laboratory material.